

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

6cTDF4hLRDTcPodcf5uAAxDrYf51DSpJV92LiDorj2Cu

Date: 5/1/2025

Compounds: Amoxicillin, Paracetamol, Cannabis, Fentanyl

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Fentamylxicar

Formula: C₂₈H₃₉N₃O₇

Weight: 517.635

Drug-likeness: 0.57

Activity:

Analgesic and antibiotic properties

Target Analysis

Opioid receptor mu (MOR)

Binding Affinity: 8.5

Crucial for pain relief management

Penicillin-binding proteins (PBPs)

Binding Affinity: 7.2

Important for antimicrobial action

Toxicity Profile

Risk Level: High

Affected Systems: Liver, Respiratory system, Gastrointestinal tract

Mitigation Suggestions:

- Dose adjustment
- Regular liver function tests
- Monitoring respiratory function

Pharmacokinetics

Absorption:

Rapid oral absorption

Distribution:

Wide distribution with high protein binding

Metabolism:

Hepatic metabolism via CYP450

Excretion:

Renal and biliary excretion

Half-life: 4-6 hours

Detailed Analysis

Fentamyxicar is designed as a hybrid compound combining the opioid receptor activity of fentanyl with the antibiotic properties of amoxicillin, further enhanced by key metabolic influences seen in cannabis and paracetamol. With a molecular weight of approximately 517 g/mol, it is specifically designed to inhibit both the mu-opioid receptor (MOR) and penicillin-binding proteins (PBPs), providing both analgesic and antimicrobial effects. Its dual-mechanism operates by preventing bacterial cell wall synthesis while effectively modulating pain perception. The compound is devised to potentially deliver broad-spectrum effects beneficial for managing comorbid conditions of pain and infection. However, it possesses high toxicity risks, notably affecting the liver and respiratory system, necessitating careful dosage regulation and patient monitoring. The synthesis methodology involves an innovative binding of active moieties to achieve synergistic effects, which in turn provides comprehensive therapeutic applications, particularly suited for patients with complex clinical presentations. The development strategy incorporates extensive pre-clinical and clinical testing phases, with a clearly outlined path towards regulatory approval by leveraging refined QSAR models and in-depth molecular biology insights.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>